

Addis Ababa Science and Technology University

College of Natural & Social Sciences

Department of Physics & Statistics

General Physics (Phy1001) Test for Freshman students

Date: July 07, 2022

Allowed Time: 1.30 Hrs.

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Instruction:

- Using Cellphone is strictly forbidden !
- Use your own calculator !
- Use the backside of the exam paper for rough calculation.

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Excellent!

Part I: Fill in the blank space with the most simplified answer(s) [6 pts.]

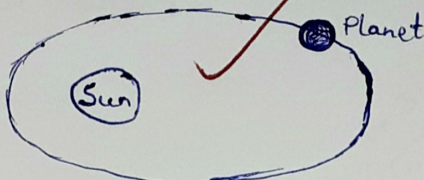
- Given two vectors : $\vec{A} = i + 2j - 2k$ and $\vec{B} = -3i - j + 4k$. Then,
A) $\vec{A} \cdot \vec{B} = \underline{-13}$
B) a vector perpendicular to both $\vec{A}$ and $\vec{B} = \underline{6i + 2j + 5k}$
C) a vector of magnitude 4 units which is perpendicular to both $\vec{A}$ and $\vec{B} = \underline{\frac{24i}{\sqrt{65}} + \frac{8j}{\sqrt{65}} + \frac{20k}{\sqrt{65}}}$
- Five objects are situated along the x and y axis as follows: a 1-kg object is at (2, 2) m, a 2-kg object is at (-2, 2) m, a 3-kg object is at (-2, -2)m, a 4-kg object is at (2, -2)m, and a 6-kg object is at (0, 0)m.
a) The X center of mass of the system = 0 m
b) The Y center of mass of the system = -0.5 m.
- An object of mass 6 kg, moving with an initial velocity of $10i$ m/s collides with another object of mass 4 kg moving with an initial velocity of $10(-i)$ m/s. If the collision is perfectly inelastic, their final velocity will be 2 m/s.

Part II: Short Answer [6 pts.]

1. State :

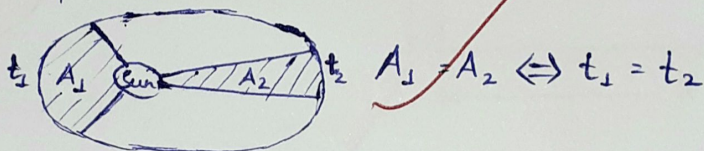
a) Kepler's 1st law (The Law of Orbits)

It states that planets revolve around the Sun in an elliptical path, with the Sun as one of the foci of the ellipse.



b) Kepler's 2nd law (The Law of Areas)

It propounds that the vector radius joining the centers of the Sun & the planet sweeps out equal areas in equal intervals of time, regardless of the distance between the Sun & the planet.



c) Kepler's 3rd law (The Law of Harmony)

It declares that the square of the time period of a planet is directly proportional to the cube of the distance joining the centers of the Sun & the planet, i.e. $T^2 \sim r^3$.

$$F_c = F_g \Rightarrow mr\omega^2 = \frac{GMm}{r^2} \quad \text{--- where } M = \text{mass of the sun} \\ m = \text{mass of the planet}$$

$$\omega^2 = \frac{GM}{r^3} \Rightarrow \left(\frac{2\pi}{T}\right)^2 = \frac{GM}{r^3} \Rightarrow \frac{4\pi^2}{T^2} = \frac{GM}{r^3} \Rightarrow \boxed{\frac{T^2}{r^3} = \frac{4\pi^2}{GM}}$$

2. State Newton's 1st law of motion (The Law of Inertia)

It states that an object tends to preserve its state of motion, that is, it continues to be at rest or move with a constant velocity unless acted upon by an external force.

Part III. Workout: Show all the necessary steps clearly & neatly [8 pts.]

1. A ball is projected upward from the edge of a cliff 100 m above the ground level with an initial speed of 20 m/s at an angle of 37° above the horizontal.

A) How long does it take the ball to reach the ground? [2 pts.]

B) What is the ball's horizontal distance X? [1 pt.]

C) What is the speed of the ball just before it strikes the ground? [2pts.]

(Use: $\sin 37 = 0.6$, $\cos 37 = 0.8$, $g = 10 \text{ m/s}^2$)

$$U = 20 \text{ m/s}, y_0 = 100 \text{ m}, \theta = 37^\circ$$

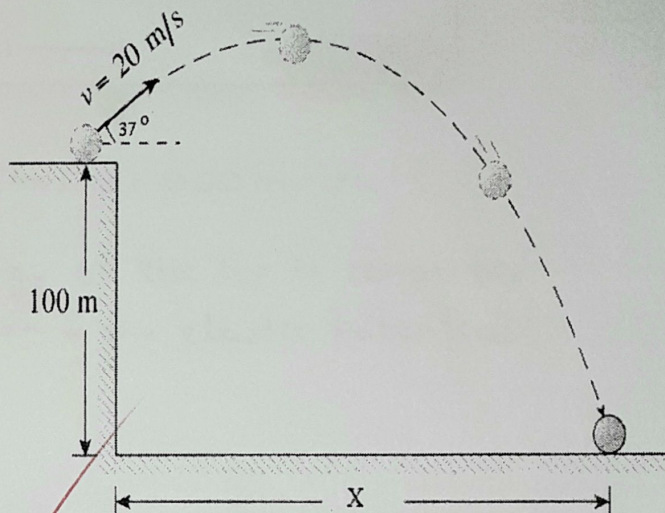
$$a) y = y_0 + U_y t - \frac{1}{2} g t^2$$

$$\frac{1}{2} g t^2 - U \sin \theta t - y_0 = 0$$

$$5t^2 - 12t - 100 = 0$$

$$t = \frac{12 + \sqrt{144 + 2000}}{10}$$

$$= 5.83 \text{ s}$$



$$b) x = U \cos \theta \cdot t = 20 (0.8) (5.83) = 93.28 \text{ m}$$

$$c) V_x = U_x = U \cos \theta = 20 (0.8) = 16 \text{ m/s}$$

$$V_y = U_y - g t = U \sin \theta - g t = 20 (0.6) - 10 (5.83) = -46.3 \text{ m/s}$$

$$V = \sqrt{V_x^2 + V_y^2} = \sqrt{16^2 + (-46.3)^2} = 49 \text{ m/s}$$

$$\beta = \tan^{-1} \left(\frac{V_y}{V_x} \right) = \tan^{-1} (-2.89) = -70.94^\circ$$

Thus, $V = 49 \text{ m/s}$ with 70.94° below the horizontal